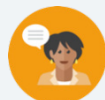


8.9 Using Key Features to Solve Problems

Lesson Introduction

 Benchmarks	MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context. MA.912.AR.3.8 Solve and graph mathematical and real-world problems that are modeled with quadratic functions. Interpret key features and determine constraints in terms of the context. MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or nonviable options.		 Learning Targets	- I can solve real-world problems modeling quadratic relationships by determining the necessary key feature and the form of the equation that reveals it. - I can interpret key features of quadratic relationships in a real-world context.	
 Benchmark Coherence	Building On		Working Toward		
	N/A		MA.912.AR.9.7 Given a real-world context, represent constraints as systems of linear and non-linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.		
 Essential Questions	- How can I interpret key features of a quadratic relationship in a real-world context? - How can I determine which key features will help me in solving a real-world problem?				
 Lesson Narrative	Students apply their learning from the entire unit to real-world problems involving quadratics. Lesson 9 incorporates a variety of elements of teaching for you to leverage based on your students: Guided Instruction, Collaboration, and more practice like 8.9.4 Your Turn! and 8.9.5 Wrap-Up.				
 Component Summary	8.9.1 Warm-Up (~5 min.)	Rewrite an equation in factored form and determine the key feature that can be determined (<i>SE p. 52</i>)	8.9.3 Collaboration (15-20 min.)	Interpret key features of a quadratic equation that represents a real-world context (<i>SE p. 54</i>)	
	8.9.2 Guided Instruction (10 min.)	Find the vertex using a strategy other than completing the square (<i>SE p. 53</i>)	8.9.4 Your Turn! (10 min.)	Rewrite quadratic equations that model a quadratic relationship in a real-world context, interpret key features (<i>SE p. 55</i>)	
	8.9.5 Wrap-Up (~5 min.)	Ticket Out the Door on real-world quadratic functions (<i>SE p. 55</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Vertex Form - Perfect Square Trinomial		N/A		This lesson will require students to read and interpret text. Consider going through this lesson independently before teaching to determine points in which you may want to model the question and others where you want to release to students to try.

 Teacher Tips	Common Misconceptions - Students may not recall how to identify the key features. - Students may not recognize a value that is not viable for a given context. - Students may misinterpret the question and the key features.	Things to Consider The problems in this lesson are quadratic functions set in the real-world. Consider refreshing students on your chosen word-problem attack strategies (see unit guide for examples). You may want to review the strategy during 8.9.2 Guided Instruction.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson deals primarily with real-world word problems. Consider leveraging the three-reads protocol for each word problem. It will support reading comprehension of problems and meta-awareness of mathematical language.	MA.K12.MTR.3.1: Mathematical Fluency Notice the connections students can make to efficiency and fluency in question 1 of 8.9.2 Guided Instruction. Learning experiences should incorporate moments for students to explore and generalize their methods, and question 1 positions students to synthesize the application of completing the square and $-\frac{b}{2a}$.
 Differentiate Instruction	Consider the progress of your students from the entire unit to inform what differentiation could look like during this lesson. 8.9.2 Guided Instruction, 8.9.3 Collaboration, or 8.9.4 Your Turn! could all be opportunities to construct stations, rotations, or small groups to provide a variety of entry points for students to synthesize their findings. - Utilize Study Expert videos from previous lessons that highlight specific concepts like completing the square. - Leverage Algebra 1 videos that highlight factoring practice for students. - Offer online graphing tools like Desmos for students to explore key features digitally.	
Lesson Notes:		

8.9.1 Warm-Up: Key Features and Forms

Component Overview: Students will rewrite an equation in factored form and determine the key feature that can be determined.



8.9.1 Warm-Up: Key Features and Forms

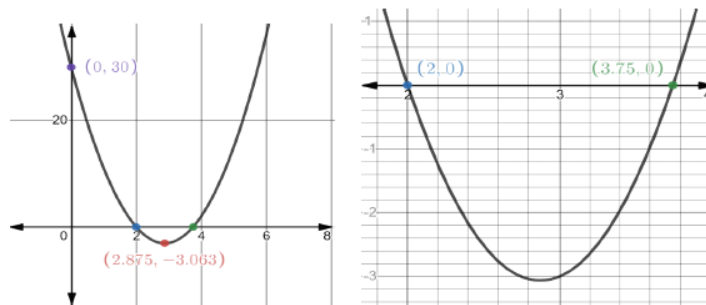
Several investors provided funds for a startup company to begin production of a new product. The company's revenue, R , in thousands of dollars, for the first year is modeled by the function $R(x) = 4x^2 - 23x + 30$ where x is the number of months since production started.

- Fill in each box with a number and each circle with an operation to rewrite $R(x)$ in factored form.

$$R(x) = (\boxed{}x \bigcirc \boxed{})(\boxed{}x \bigcirc \boxed{})$$

$$R(x) = (x - 2)(4x - 15)$$

- Two graphs of $R(x)$ are shown.



- Which key feature(s) shown on the graph can be determined from the factored form of $R(x)$?
x-intercepts
- Which key feature(s) shown on the graph can be determined from the standard form of $R(x)$?
y-intercept
- Which key feature(s) shown on the graph cannot be determined from the factored or the standard form of $R(x)$?
vertex



If students are confused by the box frames as scaffolding, prompt them to complete the square to convert the provided function into vertex form.



Would you want to determine which key features you are looking for before determining the form you would like them to be represented in? Why or why not?

8.9.2 Guided Instruction: Finding the Vertex Using Another Strategy

Component Overview: Students will receive guided instruction on finding the vertex using a strategy other than completing the square.



8.9.2 Guided Instruction: Finding the Vertex Using Another Strategy

Sometimes completing the square is not an efficient method to find the vertex, or the process becomes cumbersome. For example, the vertex for $R(x)$ in the Warm-Up is (2.875, 3.0625). On the graph in the Warm-Up, the graphing software rounded the y -value to the nearest thousandth.

The steps of rewriting $R(x) = 4x^2 - 23x + 30$ to vertex form are shown.

$$R(x) = 4\left(x^2 - \frac{23}{4}x\right) + 30$$

$$R(x) = 4\left(x^2 - \frac{23}{4}x + \frac{529}{64}\right) + 30 - \frac{529}{16}$$

$$R(x) = 4\left(x - \frac{23}{8}\right)^2 - \frac{49}{16}$$

1. Discuss with your partner how the a and b coefficients of $R(x) = 4x^2 - 23x + 30$ relate to the value of h , $\frac{23}{8}$, of the vertex form of $R(x)$.

$$\frac{-b}{2a}$$

The y -value of the vertex in the function, $R(x)$, can be found by substituting the x -value of the vertex into the quadratic equation.

2. During the first year of production for the startup company in the Warm-Up, when will the company's revenue be negative?
 $2 < x < 3.75$
3. Does it make sense for a company that's just starting to have a negative revenue?
Answers will vary. Yes, because the company may have put a lot of money out to start the company and business could be slow.



Is completing the square the most effective method here? Why or why not? (This question positions students to identify efficiency in question 1.)



Avoid language of "shortcut" or "trick" when discussing the relationship between $-\frac{b}{2a}$ and completing the square. Instead, prompt students to discuss the relationship between the two concepts and how they might leverage this relationship when identifying the vertex or converting to vertex form.



In many real-world contexts, negative values are not viable for the given context. What are some examples that a negative value could be a viable answer for a context?

8.9.3 Collaboration: Interpreting Key Features

Component Overview: Students will work together to interpret key features of a quadratic equation that represents a real-world context.



8.9.3 Collaboration: Interpreting Key Features

Work with your partner to complete the following.

- Determine the key feature that is needed to answer the question.
 - Identify the best form to use to find the key feature of the quadratic.
 - Determine the answer and show your work.
1. A scientist is performing a study on mosquitoes. She collects water with mosquito larvae and places it in an enclosed container in her lab to watch for the larvae to hatch. The scientist uses the data from the study to create the function $M(t) = -3t^2 + 57t - 144$, where t is the number of days in the study and M is number of mosquitos in the container.
- a. When did mosquitoes first appear in the container?

Key Feature Needed	Which Form Reveals This?	Answer
x -intercept	Factored form	(3,0) After 3 days.

- b. When were there are no more mosquitos in the container?

Key Feature Needed	Which Form Reveals This?	Answer
x -intercept	Factored form	(16,0) After 16 days.



For question 1, model for students a strategy that you are already using for reading math word problems. This can serve as a refresher of that process and serve students well when they work on question 2 on their own.



- Compare your answers to question 1a and question 1b. What is the same? What is different?
- Why is the key feature the same?

8.9.3 Collaboration: Interpreting Key Features (continued)

2. The function $C(x) = -2x^2 + 38x + 40$ models the sales, in hundreds of millions of dollars, of compact discs in the years since 1990.



- a. How many years did the sales of compact discs increase?

Key Feature Needed	Which Form Reveals This?	Answer
<i>Vertex</i>	<i>Vertex Form</i>	$0 < x < 9.5$ <i>Up to 9.5 years</i>

- b. When were the sales of compact discs at their greatest?

Key Feature Needed	Which Form Reveals This?	Answer
<i>Vertex</i>	<i>Vertex Form</i>	$(9.5, 220.5)$ <i>At 9.5 years</i>

- c. What was the maximum amount of sales, in millions of dollars, of compact discs?

Key Feature Needed	Which Form Reveals This?	Answer
<i>Vertex</i>	<i>Vertex Form</i>	$(9.5, 220.5)$ <i>220.5 million dollars</i>

- d. Compact discs, or CDs, were the most popular way to listen to music before music streaming became available. Discuss with your partner why the key features of the function C make sense given this context.

Answers will vary. The values make sense because they were popular at the time and it's reasonable for that many to be sold.



Based on student performance, consider using this time when students are working with their partners on question 2 to pull a small group of students, using a Think Aloud strategy to model decoding of the real-world context.



Consider extending student thinking for pairs or groups that need additional support. Students have not used the term maximum prior to question 2c. This may generate a discussion where teacher guidance may be needed.

- Is 9.5 years a reasonable value? What year does that occur?*
- Why are the vertex and vertex form used for each of the scenarios in questions 2a, 2b, and 2c?*
- What context clues helped you determine the correct key feature?*



MA.K12.MTR.6.1: Reasonableness

Continue to engage students in discussion around viability and making sense of answers within a context. Also emphasize units and interpretation of time.

8.9.4 Your Turn!

Component Overview: Students will work independently to practice rewriting quadratic equations that model a quadratic relationship in a real-world context and interpreting the key features. For this activity, release students to work on this activity on their own and review the correct answers as a class.



8.9.4 Your Turn!

- Engineers are designing a rollercoaster that goes underground at one point. The rollercoaster's path underground is a large curve. Engineers use the function shown to model the elevation, E , in feet and time, t , in seconds, of the rollercoaster's path for a portion of the ride.

$$E(t) = t^2 - 10t + 16$$

- Write an equivalent form of $E(t)$ that can be used to find the time when the rollercoaster goes underground and the time the rollercoaster returns above ground.

$$E(t) = (t - 8)(t - 2)$$

- When did the rollercoaster go underground?

At 2 seconds.

- How long was the rollercoaster underground?

6 seconds



Based on student performance, consider using this time when students are working on question 1 to pull a small group of students. This should be reserved for students who may need one-on-one support.



For an additional challenge, consider having students also rewrite the expression in vertex form, requiring them to demonstrate their progress in completing the square.

8.9.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses students' understanding of real-world quadratic functions. This activity should be done independently to get accurate student-level data. Students should be allowed to access a calculator to complete the questions.



8.9.5 Wrap-Up: Ticket Out the Door

- The equation $H(x) = -16x^2 + 96x + 6$ models the height, H , in feet, of a toy rocket x seconds after it is launched from a launchpad 6 feet above the ground.

Complete the table and show your work to determine when the rocket is at its greatest height.

Key Feature Needed	Which Form Reveals This?	Answer
<i>Vertex</i>	<i>Vertex Form</i>	<i>3 seconds</i>

$$y = -16(x^2 - 6x + 9) + 6 + 144$$

$$y = -16(x - 3)^2 + 150$$



Utilize 8.9.5 Wrap-Up to identify key features or components that can inform differentiated support as students review Unit 9.